Jagruti Thotam 23108A0066 & Samiksha Chavan 23108A0057

# SECURE NETWORK IMPLEMENTATION USING VPN, FIREWALL AND IPS

Organizations face cyber threats such as hacking, malware, and unauthorized access. A secure network is essential to protect sensitive data and maintain safe communication.

## Objective

- Design a secure network architecture
- Implement VPN for secure remote access
- Configure Firewall for traffic filtering
- Deploy IPS for attack detection and prevention

## Technologies Used

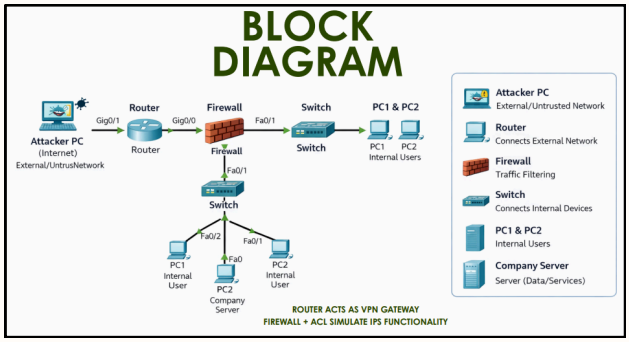
Cisco Packet Tracer

## Required Devices

- Router - 2911
- Firewall - ASA 5505
- Switches -2960
- PC1
- PC2
- Attacker_PC
- Server

## Methodology

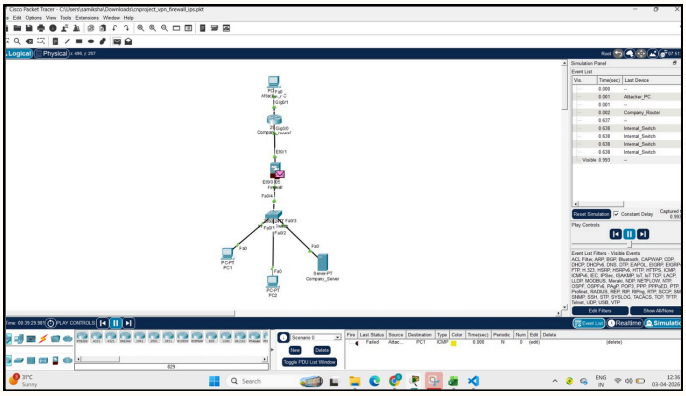
- Designed topology in Cisco Packet Tracer
- Configured IP addresses for all devices
- Connected attacker PC to external network
- Configured firewall for traffic filtering
- AppliedACL on router to simulate IPS
- Tested connectivity using ping

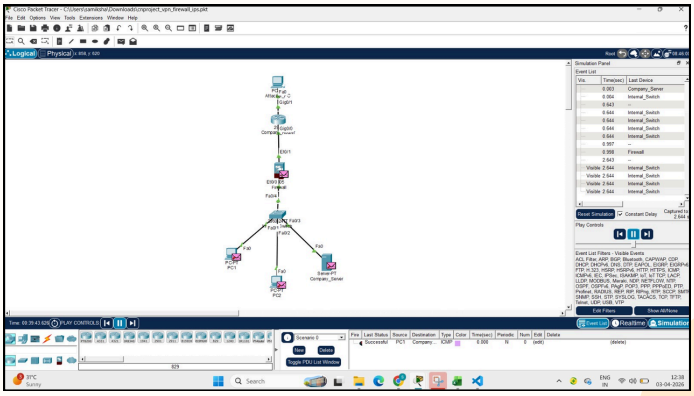



## Implementation

- ACL created to block attacker IP
- Applied ACL on router interface
- Firewall configured to filter traffic
- Router acts as VPN Gateway

## Output





## Result

- Unauthorized access blocked
- Secure communication established
- Network protected from external threats

## Conclusion

Using VPN, Firewall, and IPS together creates a strong layered security system, ensuring **confidentiality**, **integrity**, and **availability** (CIA triad) of the network.